

Released GridFM GraphKit as an AC Power-Flow Surrogate

Exact formulation of the PPC branch-row adaptation

Implementation note for `train_valid_test_gridfm.py`

August 2026

Scope. This document formulates the released `GNS_heterogeneous` from the released `gridfm_graphkit` package used with `-task pf -gridfm_impl graphkit`. It is not the earlier local `GridFMHeteroSurrogate` mirror. The optional post-model known-operator projection is excluded; all equations describe the raw GraphKit prediction.

1 AC Power-Flow Problem

Let bus voltage be

$$\underline{V}_i = v_i e^{j\theta_i}, \quad \underline{\mathbf{V}} = [\underline{V}_1, \dots, \underline{V}_N]^\top. \quad (1)$$

For the PPC-derived admittance matrix $\mathbf{Y}$, calculated nodal power is

$$\underline{\mathbf{S}}^{\text{calc}}(\mathbf{v}, \boldsymbol{\theta}) = \underline{\mathbf{V}} \odot \overline{\mathbf{Y} \underline{\mathbf{V}}}. \quad (2)$$

With specified injection $\underline{\mathbf{S}}^{\text{set}} = \mathbf{P}^{\text{set}} + j\mathbf{Q}^{\text{set}}$,

$$\Delta \mathbf{P} = \mathbf{P}^{\text{set}} - \Re(\underline{\mathbf{S}}^{\text{calc}}), \quad \Delta \mathbf{Q} = \mathbf{Q}^{\text{set}} - \Im(\underline{\mathbf{S}}^{\text{calc}}). \quad (3)$$

The PF equation masks are

$$\mathcal{M}_P = \{i : i \notin \mathcal{V}_{\text{REF}}\}, \quad \mathcal{M}_Q = \{i : i \in \mathcal{V}_{\text{PQ}}\}. \quad (4)$$

Thus active balance is imposed at PV and PQ buses, while reactive balance is imposed only at PQ buses.

2 Adapted Heterogeneous Graph

The adapted graph is

$$\mathcal{G} = (\mathcal{V}_b \cup \tilde{\mathcal{V}}_g, \mathcal{E}_{bb} \cup \mathcal{E}_{gb} \cup \mathcal{E}_{bg}). \quad (5)$$

Here $\mathcal{V}_b$ contains one node per PPC bus. In native GridFM, $\mathcal{V}_g$ contains physical generator rows. The PPC PF parquet has no generator-resolved table, so this adapter constructs one *proxy* generator $\tilde{g}_i$ for each PV or REF bus i :

$$b(\tilde{g}_i) = i, \quad |\tilde{\mathcal{G}}(i)| = \begin{cases} 1, & i \in \mathcal{V}_{\text{PV}} \cup \mathcal{V}_{\text{REF}}, \\ 0, & i \in \mathcal{V}_{\text{PQ}}. \end{cases} \quad (6)$$

The two association edges $\tilde{g}_i \rightarrow i$ and $i \rightarrow \tilde{g}_i$ do not represent electrical branches; they allow two-way latent exchange.

2.1 Bus and proxy-generator inputs

Define $T(x) = \text{sign}(x) \log(1 + |x|)$. The bus row is

$$\mathbf{x}_i^b = [T(-P_i^{\text{set}}), T(-Q_i^{\text{set}}), 0, v_i^0, \theta_i^0, \mathbb{K}_{\text{PQ}}, \mathbb{K}_{\text{PV}}, \mathbb{K}_{\text{REF}}, 0.5, 1.5, -10, 10, T(G_i^{\text{sh}}), T(B_i^{\text{sh}}), \log_{10}(V_{n,i}^{\text{KV}} + 10^{-9})] \in \mathbb{R}^{15}. \quad (7)$$

The standard PF adapter uses

$$\mathbf{x}_{\tilde{g}_i}^g = [T(\max(P_i^{\text{set}}, 0)), -10, 10, 0, 0, 0] \in \mathbb{R}^6. \quad (8)$$

The limits and costs are placeholders, not reconstructed physical-generator data. The voltage start (v^0, θ^0) is retained in the bus row.

2.2 Transformer-aware directed branch features

For branch $f \rightarrow t$, let

$$a = \tau e^{j\phi}, \quad (9)$$

and use the already per-unit branch quantities stored by the PPC loader:

$$Y_{ff} = \frac{y_f + y_f^{\text{sh}}/2}{|a|^2}, \quad Y_{ft} = -\frac{y_f}{a}, \quad (10)$$

$$Y_{tt} = y_t + y_t^{\text{sh}}/2, \quad Y_{tf} = -\frac{y_t}{a}. \quad (11)$$

Each physical branch becomes two directed message edges. Their features are

$$\mathbf{e}_{ft} = [0, 0, \Re Y_{ff}, \Im Y_{ff}, \Re Y_{ft}, \Im Y_{ft}, \tau, 0, 0, S^{\max}], \quad (12)$$

$$\mathbf{e}_{tf} = [0, 0, \Re Y_{tt}, \Im Y_{tt}, \Re Y_{tf}, \Im Y_{tf}, \tau, 0, 0, S^{\max}]. \quad (13)$$

The first two flow channels and the angle-limit channels are zero in the current PF adapter; the admittance split and tap carry the meaningful electrical data. The adapter’s reconstruction validator confirms

$$\mathbf{Y}_{\text{adapter}} = \mathbf{Y}_{\text{parquet}} \quad (14)$$

to floating-point tolerance, including bus shunts.

3 Relation-Specific Transformer Layers

Initial embeddings are produced independently by node and edge type:

$$\mathbf{h}_i^{b,(0)} = E_b(\mathbf{x}_i^b), \quad \mathbf{h}_g^{g,(0)} = E_g(\mathbf{x}_g^g), \quad \bar{\mathbf{e}}_{ji} = E_e(\mathbf{e}_{ji}), \quad (15)$$

where each E is Linear–LeakyReLU–Linear–LayerNorm. The released configuration uses hidden width $d = 48$, $H = 8$ heads, and $L = 12$ layers.

At layer ℓ , each relation has its own TransformerConv operator:

$$\mathcal{T}_{bb}^{(\ell)}, \quad \mathcal{T}_{gb}^{(\ell)}, \quad \mathcal{T}_{bg}^{(\ell)}. \quad (16)$$

For a bus edge $j \rightarrow i$ and head h , a faithful operator-level form is

$$\mathbf{q}_i^h = W_{Q,bb}^h \mathbf{h}_i^b, \quad (17)$$

$$\mathbf{k}_{ji}^h = W_{K,bb}^h \mathbf{h}_j^b + W_{K,e}^h \bar{\mathbf{e}}_{ji}, \quad (18)$$

$$\mathbf{z}_{ji}^h = W_{V,bb}^h \mathbf{h}_j^b + W_{V,e}^h \bar{\mathbf{e}}_{ji}, \quad (19)$$

$$\alpha_{ji}^h = \text{softmax}_{j \in \mathcal{N}_{bb}(i)} \left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_{ji}^h}{\sqrt{d_h}} \right), \quad (20)$$

$$\mathbf{m}_{bb,i}^{(\ell)} = W_{O,bb} \parallel \sum_{h=1}^H \alpha_{ji}^h \mathbf{z}_{ji}^h. \quad (21)$$

Association relations are analogous but have no branch attribute:

$$\mathbf{m}_{gb,i}^{(\ell)} = \mathcal{T}_{gb}^{(\ell)}(\mathbf{h}_i^b, \{\mathbf{h}_g^g : b(g) = i\}), \quad (22)$$

$$\mathbf{m}_{bg,g}^{(\ell)} = \mathcal{T}_{bg}^{(\ell)}(\mathbf{h}_g^g, \mathbf{h}_{b(g)}^b). \quad (23)$$

HeteroConv sums relation outputs at each target type, then applies LayerNorm, LeakyReLU, and a shape-compatible residual connection:

$$\mathbf{h}_i^{b,(\ell+1)} \leftarrow \mathbf{h}_i^{b,(\ell)} + \sigma \left(\text{LN}(\mathbf{m}_{bb,i}^{(\ell)} + \mathbf{m}_{gb,i}^{(\ell)}) \right), \quad (24)$$

$$\mathbf{h}_g^{g,(\ell+1)} \leftarrow \mathbf{h}_g^{g,(\ell)} + \sigma \left(\text{LN}(\mathbf{m}_{bg,g}^{(\ell)}) \right). \quad (25)$$

The first layer changes width from 48 to $48H$, so its outer residual is omitted; subsequent layers retain it. TransformerConv also uses its internal β -gated root/neighbor combination.

4 Layerwise Physics Feedback

Unlike the local mirror, the released model invokes the same bus and generator decoders after *every* layer:

$$[\tilde{v}_i^{(\ell)}, \tilde{\theta}_i^{(\ell)}] = D_b(\mathbf{h}_i^{b,(\ell)}), \quad \tilde{P}_g^{(\ell)} = D_g(\mathbf{h}_g^{g,(\ell)}). \quad (26)$$

PF masks pin known state components:

$$v_i^{(\ell)} = \begin{cases} \tilde{v}_i^{(\ell)}, & i \in \mathcal{V}_{\text{PQ}}, \\ v_i^0, & i \in \mathcal{V}_{\text{PV}} \cup \mathcal{V}_{\text{REF}}, \end{cases} \quad (27)$$

$$\theta_i^{(\ell)} = \begin{cases} \theta_i^0, & i \in \mathcal{V}_{\text{REF}}, \\ \tilde{\theta}_i^{(\ell)}, & \text{otherwise.} \end{cases} \quad (28)$$

For a directed edge $i \rightarrow j$, the branch-current and branch-power operators are

$$\underline{I}_{ij}^{(\ell)} = Y_{ii}^{(ij)} \underline{V}_i^{(\ell)} + Y_{ij}^{(ij)} \underline{V}_j^{(\ell)}, \quad (29)$$

$$\underline{S}_{ij}^{(\ell)} = \underline{V}_i^{(\ell)} \overline{\underline{I}_{ij}^{(\ell)}}. \quad (30)$$

Outgoing directed powers are aggregated at each bus:

$$P_i^{\text{net},(\ell)} = \sum_{j:(i,j) \in \mathcal{E}_{bb}} P_{ij}^{(\ell)}, \quad Q_i^{\text{net},(\ell)} = \sum_{j:(i,j) \in \mathcal{E}_{bb}} Q_{ij}^{(\ell)}. \quad (31)$$

The PF physics decoder analytically recovers quantities that are not specified:

$$Q_{g,i}^{(\ell)} = \begin{cases} Q_i^{\text{net},(\ell)} + Q_{d,i} - q_i^{\text{sh},(\ell)}, & i \in \mathcal{V}_{\text{PV}} \cup \mathcal{V}_{\text{REF}}, \\ 0, & i \in \mathcal{V}_{\text{PQ}}, \end{cases} \quad (32)$$

$$P_{g,i}^{(\ell)} = \begin{cases} \sum_{g:b(g)=i} \tilde{P}_g^{(\ell)}, & i \in \mathcal{V}_{\text{PV}}, \\ P_i^{\text{net},(\ell)} + P_{d,i} - p_i^{\text{sh},(\ell)}, & i \in \mathcal{V}_{\text{REF}}, \\ 0, & i \in \mathcal{V}_{\text{PQ}}. \end{cases} \quad (33)$$

With $p_i^{\text{sh}} = -G_i^{\text{sh}} v_i^2$ and $q_i^{\text{sh}} = B_i^{\text{sh}} v_i^2$, the layer residual is

$$r_{P,i}^{(\ell)} = P_{g,i}^{(\ell)} - P_{d,i} + p_i^{\text{sh},(\ell)} - P_i^{\text{net},(\ell)}, \quad (34)$$

$$r_{Q,i}^{(\ell)} = Q_{g,i}^{(\ell)} - Q_{d,i} + q_i^{\text{sh},(\ell)} - Q_i^{\text{net},(\ell)}. \quad (35)$$

It is injected into the next bus representation:

$$\mathbf{h}_i^{b,(\ell)} \leftarrow \mathbf{h}_i^{b,(\ell)} + \text{MLP}_{\text{phys}}([r_{P,i}^{(\ell)}, r_{Q,i}^{(\ell)}]). \quad (36)$$

Only the last layer's decoded state is returned. This is depthwise learned physics feedback, not a Newton step and not a K -step voltage unrolling.

5 Output and Training Objective

The raw AC-PF prediction is

$$\hat{\mathbf{V}}_{\text{state}} = \{(\hat{v}_i, \hat{\theta}_i) = (v_i^{(L)}, \text{wrap}(\theta_i^{(L)}))\}_{i=1}^N. \quad (37)$$

Against Newton labels (v_i^*, θ_i^*) , the supervised term is

$$\mathcal{L}_{\text{state}} = \frac{1}{N} \sum_i \left[(\hat{v}_i - v_i^*)^2 + \text{wrap}(\hat{\theta}_i - \theta_i^*)^2 \right]. \quad (38)$$

The external PPC physics term recomputes power from the exact dataset $\mathbf{Y}$, not from transformed neural features:

$$\mathcal{L}_{\text{phys}} = \frac{1}{|\mathcal{M}_P| + |\mathcal{M}_Q|} \left[\sum_{i \in \mathcal{M}_P} \rho(\Delta P_i) + \sum_{i \in \mathcal{M}_Q} \rho(\Delta Q_i) \right], \quad (39)$$

where the default robust penalty is

$$\rho(r) = \log \cosh(\text{clip}(r, -30, 30)). \quad (40)$$

The default total objective is

$$\mathcal{L} = \mathcal{L}_{\text{state}} + 10^{-2} \mathcal{L}_{\text{phys}}. \quad (41)$$

Training physics is evaluated through the script’s complex64 path; reported validation/test residual metrics reconstruct $\underline{\mathbf{S}}^{\text{calc}}$ with complex128 when the dataset is loaded with `-dataset_complex_dtype complex128`.

6 Interpretive Summary

Property	Released GraphKit PF adaptation
Graph	Bus nodes plus one proxy generator per PV/REF bus; three directed relation types.
Electrical edge encoding	Directed transformer-aware π -model split (Y_{ff}, Y_{ft}) and (Y_{tt}, Y_{tf}) .
Message passing	12 relation-specific multi-head TransformerConv layers, width 48 and 8 heads.
Voltage decoding	Shared decoder at every layer; known PF quantities are pinned.
Physics inside forward	Branch flows, nodal injections, analytic PF decoder, and residual feedback at every layer.
Physics outside forward	Exact-Ybus robust residual loss and complex128 validation/test metrics.
Pretraining status	No released GridFM checkpoint; this implementation is trained from scratch.

Code provenance

The equations follow `gridfm_graphkit/models/gnn_heterogeneous_gns.py`, `gridfm_graphkit/models/utils.py`, `gridfm_graphkit_adapter.py`, and the AC-PF path in `train_valid_test_gridfm.py`, as present in the local workspace in August 2026.